MATH2601 – Higher Linear Algebra

Topic 5 – Determinants

Lecture 5.2 – Determinants in terms of other matrices

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Elementary matrices

Recall that in the process of Gaussian elimination, there are three standard types of elementary row operations that can be performed to a matrix without changing the solution set to the system of equations it represents:

- Swap two rows.
- Multiply a row by a nonzero constant.
- Add to one row the multiple of another row.

Notice that each operation when applied to a matrix $A \in M_{m,n}(\mathbb{F})$ can also be represented as premultiplication by some matrix $E \in M_{m,m}(\mathbb{F})$ found by applying the same row operation to the identity matrix $\mathcal{I}_m$. Writing $\mathcal{I}_m = \begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \dots & \mathbf{e}_m \end{pmatrix}$, we have that premultiplying A by...

- $E_1 = \begin{pmatrix} \mathbf{e}_1 & \dots & \mathbf{e}_j & \dots & \mathbf{e}_i & \dots & \mathbf{e}_m \end{pmatrix}$ swaps the ith and jth rows (for $i < j$).
- $E_2 = \begin{pmatrix} \mathbf{e}_1 & \dots & \lambda\mathbf{e}_i & \dots & \mathbf{e}_m \end{pmatrix}$ multiplies the ith row by λ.
- $E_3 = \begin{pmatrix} \mathbf{e}_1 & \dots & \lambda\mathbf{e}_i + \mathbf{e}_j & \dots & \mathbf{e}_m \end{pmatrix}$ adds λ times the jth row to the ith row.

Notice that $\det(E_1) = -1$, $\det(E_2) = \lambda$, and $\det(E_3) = 1$. This matches the change to the determinant of any square A after each row operation (see Lecture 5.1). So for each i, we have $\det(E_iA) = \det(E_i)\det(A)$.

Lemma. Given $A \in M_{m,n}(\mathbb{F})$ and any elementary matrix $E \in M_{m,m}(\mathbb{F})$, we have $\det(EA) = \det(E)\det(A)$.

Matrix inverses

Recall from first year (or Lecture 3.2) that the method to find the inverse of a matrix $A \in M_{m,n}(\mathbb{F})$ is to row-reduce the matrix $(\ A \mid I\ )$ until the matrix left of the bar is I, in which case A^{-1} is the matrix right of the bar.

Lemma. Given a matrix $A \in \mathrm{GL}(n, \mathbb{F})$, there exist elementary matrices $E_1, E_2, \dots, E_k$ such that $A = E_1 E_2 \dots E_k$ (so $A^{-1} = E_k^{-1} \dots E_2^{-1} E_1^{-1}$) and $\det(A) = \det(E_1)\det(E_2)\dots\det(E_k)$.

Proof. Using the method of finding inverses described above, the process of row-reducing A to I translates to the matrix multiplication $F_k F_{k-1} \dots F_2 F_1 A = I$ for elementary matrices F_i. Note that for each i, the matrix F_i^{-1} is an elementary matrix since the reversal of an elementary row operation is itself an elementary row operation. Writing $E_i = F_i^{-1}$ for each i, rearrangement gives $A = E_1 E_2 \dots E_k$. Then from the previous lemma, $\det(A) = \det(E_1)\det(E_2)\dots\det(E_k)$.

Invertibility

Lemma. A matrix $A \in M_{n,n}(\mathbb{F})$ is invertible if and only if $\det(A) \neq 0$.

Proof. For the forward direction, by the previous lemma, if A is invertible then $\det(A) = \det(E_1)\det(E_2)\ldots\det(E_k)$ for some elementary matrices E_i, so since no elementary matrix has determinant 0, nor does A.

For the other direction, if A is not invertible then its columns are not linearly independent, meaning one of the columns (say column i) can be written as a linear combination of the others. Elementary row operations can then be applied to A^T to replace row i with a row of zeros. But this matrix has determinant 0, implying $\det(A) = \det(A^T) = 0$.

Multiplicativity of determinant

Theorem. Given $A, B \in M_{n,n}(\mathbb{F})$, we have $\det(AB) = \det(A)\det(B)$.

Proof. Suppose A is invertible, so that it can be written as the product of elementary matrices $E_1, E_2, \ldots, E_k$. Then clearly

$$\det(AB) = \det(E_1 \ldots E_k B) = \det(E_1) \ldots \det(E_k)\det(B) = \det(A)\det(B).$$

Otherwise, suppose A is not invertible so that $\det(A) = 0$. If AB were invertible, there would be some matrix C such that $(AB)C = A(BC) = I$. But then A would have BC as an inverse, contradicting the assumption. So AB is also not invertible, meaning $\det(AB) = 0 = \det(A)$.

Corollary. For any $A \in \mathrm{GL}(n, \mathbb{F})$ we have $\det(A^{-1}) = \det(A)^{-1}$.

Proof. Since $AA^{-1} = I$, taking the determinant of both sides gives $\det(A)\det(A^{-1}) = \det(I) = 1$, so $\det(A^{-1}) = \det(A)^{-1}$.

Corollary. The determinant is a matrix similarity invariant.

Proof. If $A, B \in M_{n,n}(\mathbb{F})$ are similar matrices, then $B = P^{-1}AP$ for some $P \in \mathrm{GL}(n, \mathbb{F})$. Taking determinants of both sides gives

$$\begin{aligned}\det(B) = \det(P^{-1}AP) &= \det(P^{-1})\det(A)\det(P) \\ &= \det(P)^{-1}\det(A)\det(P) = \det(A).\end{aligned}$$

Another multiplicativity proof

Lemma. Let $\mathbb{F}^{n\times n}$ be the direct product of n copies of $\mathbb{F}^n$. The only mapping $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ that is multilinear, alternating, and for which $f(I) = 1$ is the determinant.

Proof. Let $A \in M_{n,n}(\mathbb{F})$ and suppose $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ has the listed properties. Writing $\mathbf{e}_i$ as the ith standard basis vector in $\mathbb{F}^n$, we can write the jth column of A as $\sum_{i=1}^n a_{i,j}\mathbf{e}_i$. So applying f to the columns of A gives

$$\begin{aligned} f(A) &= f\Big(\sum_{i_1=1}^n a_{i_1,1}\mathbf{e}_{i_1}, \sum_{i_2=1}^n a_{i_2,2}\mathbf{e}_{i_2}, \dots, \sum_{i_n=1}^n a_{i_n,n}\mathbf{e}_{i_n}\Big) \\ &= \sum_{i_1=1}^n \sum_{i_2=1}^n \cdots \sum_{i_n=1}^n a_{i_1,1} a_{i_2,2} \dots a_{i_n,n}\, f(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, \dots, \mathbf{e}_{i_n}), \end{aligned}$$

since f is multilinear. Notice that if $i_j = i_k$ for any $j \neq k$, then $f(\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_n})$ should take the same value when $\mathbf{e}_{i_j}$ and $\mathbf{e}_{i_k}$ are swapped, but since f is alternating these two values should be each other's negative – so in all such cases the value is 0.

So the only nonzero summands will be those for which there are no repeated $\mathbf{e}_i$ terms, meaning they must be a permutation of the standard basis.

Another multiplicativity proof

Lemma. Let $\mathbb{F}^{n\times n}$ be the direct product of n copies of $\mathbb{F}^n$. The only mapping $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ that is multilinear, alternating, and for which $f(I) = 1$ is the determinant.

Proof (continued). Thus

$$f(A) = \sum_{\sigma\in S_n} \prod_{i=1}^{n} a_{\sigma(i),i}\, f(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}) = \sum_{\sigma\in S_n} \prod_{i=1}^{n} a_{\sigma(i),i},$$

since $f(I) = 1$ by assumption. But this is exactly the formula for $\det(A^T) = \det(A)$.

Theorem. Given $A, B \in M_{n,n}(\mathbb{F})$, we have $\det(AB) = \det(A)\det(B)$.

Proof. Fix $A \in M_{n,n}(\mathbb{F})$ and let f be the function given by $f(B) = \det(AB)$. It can be shown that f is multilinear and alternating (prove this!). Furthermore $f(I) = \det(A)$, so $\det(A)^{-1} f(I) = 1$, meaning by the previous lemma that in general $\det(A)^{-1} f(B) = \det(B)$ and so $f(B) = \det(A)\det(B)$.

Cofactor expansion

In Lecture 5.1, we showed that in a matrix $A \in M_{n,n}(\mathbb{F})$, if the entry $a_{i,j}$ is the only nonzero entry in the ith row, then $\det(A) = (-1)^{i+j} a_{i,j} \det(A_{i,j})$ where $A_{i,j}$ is the (i,j)-minor of A.

Using the multilinearity of the determinant, we get the stronger result:

Lemma. For any $A \in M_{n,n}(\mathbb{F})$, we have for any fixed j that

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{i,j} \det(A_{i,j}),$$

and likewise for indexing over j for any fixed i.

This is known as cofactor expansion due to the below definition.

Definition. The cofactor of any entry $a_{i,j}$ in $A \in M_{n,n}(\mathbb{F})$ is the number given by $c_{i,j} = (-1)^{i+j} \det(A_{i,j})$.

Using this notation, we have that $\det(A) = \sum_{i=1}^{n} a_{i,j} c_{i,j}$ for any fixed j (and likewise for indexing over j for any fixed i).

Adjugate matrix

Definition. The adjugate (or adjunct) matrix of $A \in M_{n,n}(\mathbb{F})$, denoted $\mathrm{adj}(A)$, is the transpose of the matrix of corresponding cofactors. That is, $\mathrm{adj}(A)$ is the matrix with (i,j)th entry $c_{j,i} = (-1)^{i+j}\det(A_{j,i})$.

Lemma. For any $A \in \mathrm{GL}(n,\mathbb{F})$, we have $A^{-1} = \frac{1}{\det(A)}\,\mathrm{adj}(A)$.

Proof. Notice that the (i,j)th entry of $\mathrm{adj}(A)A$ is given by $\displaystyle\sum_{k=1}^{n} a_{k,i}c_{k,j}$.

Clearly if $i = j$, this is $\det(A)$. Otherwise if $i \neq j$. this value is $\det(A')$ for some A' whose jth column is equal to its ith column. But the determinant of any matrix with two equal columns is 0. So altogether, we find that $\mathrm{adj}(A)A = \det(A)I$, from which the result follows.

Example. The adjugate of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\mathrm{adj}(A) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}^T$, and so $A^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.